

Q. 1 Let a, b be distinct natural numbers such that their product ab is a perfect square. Prove that if $a + b$ is prime, then it must be of the form $4k + 1$, where k is some natural number.

Q. 2 Let

$$f(x) = a_0 + a_1x + \cdots + a_nx^n$$

be a polynomial whose coefficients satisfy the conditions $0 \leq a_i \leq a_0$ ($i = 0, 1, \dots, n$). Let

$$(f(x))^2 = b_0 + b_1x + \cdots + b_{n+1}x^{n+1} + \cdots + b_{2n}x^{2n}.$$

Prove that

$$b_{n+1} \leq \frac{1}{2}(f(1))^2.$$

Q. 3 Suppose that $\triangle ABC$ is right angled with $\angle ABC = 90^\circ$. Let the angle bisector of $\angle ABC$ meet AC at P . If $AP^2 = \text{area}(\triangle ABC)$, prove that $\triangle ABC$ is isosceles.

Q. 4 Let $f : [0, \pi] \rightarrow \mathbb{R}$ be continuous. Prove that

$$\int_0^\pi f(x) \sin(x) dx \leq \int_0^\pi f(x)^2 dx + \frac{\pi}{8}.$$

Q. 5 Suppose that $n > 1$ is a natural number which is not a power of 2. Prove that $\binom{2n}{n}$ must be divisible by 4.

Q. 6 Suppose that a, b, p, q, r, s are positive integers for which

$$\frac{p}{q} < \frac{a}{b} < \frac{r}{s},$$

and $qr - ps = 1$. Prove that $b \geq q + s$.

Q. 7 Assume that the quadratic polynomial $P(x) = x^2 + ax + b$ has integer roots, and there exists an integer n such that $P(n) = 13$. Prove that either $P(n + 1)$ or $P(n - 1)$ is equal to 28.

Q. 8 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function, differentiable on $(0, 1)$, with $f(0) = 0$ and $f(1) = 1$. Prove that for any natural number n , there exist n distinct points $c_1, c_2, \dots, c_n \in (0, 1)$ such that

$$\sum_{i=1}^n \frac{1}{f'(c_i)} = n.$$

-End of the Question Paper-